CausalCache: Conditional Marginal Utility of Restoring Visual History for Long-Horizon GUI Agents

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Abstract

Long-horizon GUI agents keep compressed textual summaries of past events and a bounded number of high-fidelity history screenshots. Under a fixed visual-memory budget B, the standard Recent-B policy allocates every slot to the most recent observations. We introduce CausalCache, which instead selects at most B events from the *full* interaction history—recent frames included—so its benefit comes from *reallocating* a bounded memory budget rather than adding extra visual context. The number of selected non-recent events k is a measured behavioral statistic, not a preset parameter. Its history-gated key/value (HGKV) adapter adds low-rank residuals only to restored history-image tokens and is structurally bypassed with no history image, exactly preserving the frozen policy. It is trained on matched-budget *replacement* groups—Recent-B versus the same window with its oldest slot replaced by one task-relevant archived frame—with a difference-in-differences objective whose per-arm frozen anchors make "amplify any history image" worth exactly zero. At B=1 on desktop decision groups from successful AgentNet trajectories, the frozen policy shows no reliable preference for the target-relevant archived frame over the true previous frame ($+0.001$ nats/token, 95% CI $[-0.024, +0.025]$): selectivity is not inherited. HGKV creates it ($+0.010$ $[+0.006, +0.014]$) within a pre-registered drift envelope, while a full-layer LoRA control attains selectivity only by drifting beyond it. A budget-aware selector then chooses at most B events from the full history with STOP allowed. **[Pending: Insert per-B replacement gates, the $Q_B(k)$ reallocation curve, and zero-shot MobileWorld and OSWorld closed-loop results.]**

Introduction

GUI agents accumulate screenshots, actions, text arguments, coordinates, and interface changes over many decisions. Retaining every screenshot is expensive and can distract a multimodal policy; text-only trajectories discard toggle state, layout, transient values, and spatial correspondence. A practical agent must therefore decide which historical events remain available in high fidelity (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026).

Most memory controllers frame this as unconditional selection: retain recent events, retrieve visually similar screenshots, or learn salience from task supervision (Zeng et al. 2026; Shi et al. 2026; Liu et al. 2026). Two things are missing from that framing. First, the value of an archived screenshot depends on what visual context is already present: with no recent images a restored old frame may lack local grounding, while with a saturated recent window any additional image may be redundant. Second, the action policy must actually consume restored evidence; our experiments show that a strong frozen policy can respond to the *presence* of history images without being reliably sensitive to their *content*, in which case a better selector has nothing to improve.

CausalCache therefore poses the memory question as *budget reallocation*. Under a fixed budget of B high-fidelity history images, Recent-B is one particular allocation—all slots to the most recent frames. CausalCache selects at most B events from the full history, with recent frames themselves as candidates and STOP allowed; the comparison $Q(S^*) > Q(S_{\text{recent}})$ at matched B isolates allocation quality from visual capacity. The number of selected non-recent events, $k = |S \setminus \text{Recent-}B|$, is a *result* statistic. The reallocation curve $Q_B(k)$ for $k = 0, \ldots, B$ is genuinely empirical: all recent may over-weight recency, all archived may lose local continuity, and a small replacement may be best—an inverted U that cannot be assumed in advance. We train and gate at $B \in \{1, 2, 4\}$ and evaluate B=8 only as an out-of-training extension.

Learning to *use* restored evidence must not corrupt the base agent. Full-layer adaptation simultaneously changes current-screen understanding, action syntax, and grounding; we show empirically that it gains selectivity while drifting beyond a pre-registered envelope on recent-only and wrong-history prompts. CausalCache instead trains a history-gated KV interface (HGKV) that touches only restored history-image tokens and is structurally bypassed at zero budget, with a difference-in-differences (DiD) objective in which every arm is anchored to its own frozen score: uniform amplification of any history cancels exactly, and only *selective* use of target-relevant content reduces the loss.

Our contributions are:

- a fixed-budget reallocation formulation of GUI visual memory: Recent-B versus at-most-B selection from the full history at matched image counts, with the replacement intensity k as a measured statistic and $Q_B(k)$ as the reallocation curve;

- a policy-preserving HGKV interface trained on matched-budget replacement groups with per-arm-anchored DiD supervision, exact no-history parity, and pre-registered caps on recent-window and wrong-history drift;
- matched structural controls showing that full-layer adaptation attains selectivity only outside the drift envelope, and that token gating adds selective gain beyond a layer- and rank-matched ungated control;
- a budget-aware selector over the full candidate history (recent frames included, STOP allowed), trained on desktop data only and evaluated zero-shot on mobile and desktop benchmarks.

Related Work

Visual GUI agents and benchmarks. Screenshot-based GUI agents increasingly learn grounding and action prediction directly from pixels, including CogAgent, SeeClick, ShowUI, and GUI-Owl (Hong et al. 2024; Cheng et al. 2024; Lin et al. 2025; Xu et al. 2026). AndroidWorld provides dynamically instantiated mobile tasks with executable success checks (Rawles et al. 2025), and OSWorld evaluates agents in real desktop applications (Xie et al. 2024). GUI-Odyssey supplies cross-application mobile trajectories (Lu et al. 2025); our early mobile experiments on it motivated the conditional formulation used here. We use GUI-Owl-1.5-8B as the frozen cross-platform base policy (Xu et al. 2026), train on desktop trajectories from AgentNet-style successful demonstrations, and evaluate zero-shot on held-out desktop tasks and a cross-app mobile memory benchmark. **[Pending: Cite the MobileWorld benchmark report once its canonical reference is fixed.]**

Long-horizon GUI memory. GUI-Odyssey couples cross-app trajectories with a history resampler that attends to prior screenshots (Lu et al. 2025). MementoGUI learns multimodal memory writing, compression, and retrieval operators (Zeng et al. 2026). AndroTMem retrieves causally linked anchor states from interaction histories (Shi et al. 2026). These systems establish that long-horizon GUI control benefits from selective memory, but score candidates largely independently of how much recent visual context is already present. CausalCache makes that conditioning explicit and additionally trains the policy-side interface that consumes the restored evidence.

Interventional memory value. Recent work studies memory through controlled interventions or active state influence (Srivastava 2026; Liu et al. 2026). Our intervention is narrower and executable: every event always retains the same low-fidelity record, while a selected event additionally exposes exactly one archived post-action image, holding the remaining prompt fixed. The resulting score is a policy-specific behavioral surrogate rather than an environment-level causal effect. Coalition-conditioned marginals are related to cooperative-game attribution (Shapley 1953), but our goal is a deployable budgeted selector for a fixed GUI policy, not a generic attribution value.

Problem Formulation

At decision t, the agent receives instruction g, current observation x_t, and history

$$H_t = \{e_1, \dots, e_{t-1}\}, \qquad e_j = (x_j, a_j, x_{j+1}), \quad (1)$$

where each event keeps a compact low-fidelity record $\tilde{e}_j$ and an archived high-fidelity post-action screenshot v_j.

Fixed-budget allocation. Every prompt contains the goal g, the *complete* textual action summaries $\{\tilde{e}_1, \dots, \tilde{e}_{t-1}\}$, and the current screenshot x_t. Under a budget of B high-fidelity history images, an allocation is a set $S \subseteq \{1, \dots, t-1\}$ of restored events with $|S| \leq B$. Let $Q(S)$ denote the policy's mean target-token log-likelihood of the successful next action under allocation S. The standard baseline is

$$S_{\text{recent}} = \text{Recent-}B, \quad (2)$$

the B most recent distinct history frames. CausalCache targets

$$S^* = \arg \max_{\substack{S \subseteq \{1,\dots,t-1\} \\ |S| \leq B}} Q(S), \qquad Q(S^*) > Q(S_{\text{recent}}), \quad (3)$$

so recency is one candidate allocation, not a separate modality. The replacement intensity

$$k = |S \setminus \text{Recent-}B| \quad (4)$$

is a measured statistic, and the reallocation curve

$$Q_B(k) = \max_{\substack{|S| \leq B \\ |S \setminus \text{Recent-}B| = k}} Q(S), \qquad k = 0, \dots, B, \quad (5)$$

localizes where reallocation pays. All comparisons at a given B hold the image count, resolutions, summaries, and prompt structure fixed; only which events occupy the slots differs. Cross-B comparisons are analysis, not the causal contrast. Given a partial allocation S, conditional marginals $\Delta_t(j \mid S) = Q(S \cup \{j\}) - Q(S)$ drive selection, and a replacement is justified only when the incoming event's marginal exceeds that of the recent frame it displaces. Every formally reported set utility reruns the complete selected set; we never report a sum of singleton gains as the utility of a coalition.

CausalCache

Policy-Preserving History-Gated KV Adapter

Let h_l be the input to layer l's attention projections. In the last eight language-model layers, CausalCache adds rank-8, α=16 low-rank residuals (Hu et al. 2022) only to key and value projections:

$$\begin{aligned} K_l &= W_l^K h_l + M_{\text{hist}}\, \Delta W_l^K h_l, \\ V_l &= W_l^V h_l + M_{\text{hist}}\, \Delta W_l^V h_l. \end{aligned} \quad (6)$$

M_{hist} is a token mask that is true only for restored historical image tokens. It excludes instruction tokens, event-summary text, the current observation image, and action tokens. The mask is constructed fail-closed from the multimodal token-type sequence and per-image patch geometry; any mismatch between declared images and encoded image blocks aborts the forward pass. When no history image is restored, the adapter context is absent and the projection hook returns the original module output, so $\pi_{\text{HG}}(\cdot \mid S{=}\varnothing) = \pi_0(\cdot \mid S{=}\varnothing)$ up to bitwise equality in our implementation.

Per-B Replacement Supervision

Training states are decision points of successful desktop trajectories in which the target action a_t^* recurs earlier in the same trajectory under full-action equivalence (type, arguments, and coordinates within tolerance); the screenshot preceding the earlier occurrence is the target-specific positive frame. For each state and each budget B we construct three matched-budget allocations with identical summaries, current screen, and target—every prompt carries exactly B history images:

- **Recent**: $\text{Recent-}B$, the B most recent distinct frames;
- **Sparse**: $\text{Recent-}(B-1) + \{v_{j+}\}$, the oldest recent slot replaced by the target-specific archived frame ($\text{age} \geq B+2$, strictly older than the whole window);
- **Wrong**: the same slot replaced by an age-matched same-trajectory frame whose following action is *not* equivalent to a_t^*.

Training teaches the $k=1$ replacement interface; larger-k compositions arise at selection time. Positives do not require the base action to be wrong: both information addition and evidence amplification are admissible mechanisms.

Let ℓ denote mean target log-likelihood, superscripted by adapter state (active or frozen bypass). Each arm is anchored to *its own* frozen score:

$$
\begin{aligned}
A_s &= \ell^{\text{on}}_{\text{sparse}} - \ell^{\text{off}}_{\text{sparse}}, \qquad A_r = \ell^{\text{on}}_{\text{recent}} - \ell^{\text{off}}_{\text{recent}}, \\
A_n &= \ell^{\text{on}}_{\text{wrong}} - \ell^{\text{off}}_{\text{wrong}},
\end{aligned} \tag{7}
$$

and the DiD objective is

$$
\begin{aligned}
\mathcal{L} = {} & [m - (A_s - A_r)]_+ + [m - A_s]_+ \\
& + [m - (A_s - A_n)]_+ \\
& + \lambda_{\text{cap}}\big([|A_r| - \epsilon]_+ + [|A_n| - \epsilon]_+\big) + \lambda \|\Delta W\|_2^2,
\end{aligned} \tag{8}
$$

with $m=0.01$, $\epsilon=0.02$, $\lambda_{\text{cap}}=2$, $\lambda=10^{-4}$, and no action cross-entropy. On the identity adapter every increment is zero, so $A_s - A_r$ is exactly zero: the frozen policy's own preference for target-relevant frames cannot masquerade as adapter skill, and uniformly amplifying all history buys nothing. The dead-zone caps bound recent-window and wrong-history drift with gradients on the same scale as the selection hinges. Losses and evaluation are stratified by B; a pooled mean could hide a method that helps at $B=4$ but fails at $B=1$.

Success-Anchored Utility and Selector Supervision

After the history interface is frozen, selector labels are true policy utilities: for each state and budget, singleton utilities for the full candidate pool (recent frames included), the matched $Q(\text{Recent-}B)$ anchor, and—for budgets beyond one—set utilities along shortlisted, recent, greedy, and small-beam paths. All labels rerun complete prompts through the frozen HGKV policy.

Budget-Aware Selector over the Full History

The selector should use the same interface through which the policy reads history. For each candidate and each adapted layer we capture a lightweight counterfactual readout: the frozen decision query attends to the candidate's history keys/values with and without the adapter increments, and the mean-head difference plus per-head norms give 160 features per layer (1,280 per candidate). The input additionally encodes, explicitly, candidate age and recency-window membership, action family, summary features, and the remaining budget.

A shared encoder with learned attention over the eight adapted layers feeds four heads: predicted conditional marginal $\widehat{\Delta}_t(j \mid S)$, within-state rank, positive probability, and STOP. Set-conditioned selection for $B>1$ contextualizes selected-event embeddings with one self-attention block and maps remaining candidates to that context with one cross-attention block (Zaheer et al. 2017; Lee et al. 2019). Recent frames are ordinary candidates: at each step the best remaining candidate rank is compared against STOP, and a replacement of a recent frame happens only when an archived event outranks it, realizing Eq. 3. Selector labels cover $B \in \{1, 2, 4\}$; the realized replacement intensity k (Eq. 4) is reported per budget, never preset.

Experimental Setup

Training Data and Model Selection

Policy adaptation and selector training use only desktop data: successful AgentNet/OpenCUA Ubuntu trajectories (5,000 screened, 2,293 successful, 6,003 decision points) yield fixed-budget replacement groups at $B=1/2/4$ of $969/764/470$ (2,203 in total; 1,774 training units and 211 development groups, trajectory-disjoint; click 52–56%, plus key/type/drag/scroll), complemented by a high-precision OSWorld witness seed mined from official successful trajectories. A further 160 groups at $B=8$ are constructed but deliberately excluded from training, so $B=8$ evaluates out-of-training budget extrapolation. Checkpoints and thresholds are selected only on the desktop development split under pre-registered gates: positive DiD selection with a positive bootstrap lower bound, and drift caps $|A_r|, |A_n| < \epsilon$. No mobile benchmark data touches training or selection.

The Fixed-Budget Replacement Design

Table 1 defines the core comparison. Each row fixes the budget B; every compared allocation carries exactly B history images with identical resolutions, summaries, and prompt structure, so the contrast isolates *which events occupy the slots*. $B=4$ is the primary deployment-like setting.

Policies and Memory Baselines

The frozen policy is GUI-Owl-1.5-8B-Instruct (Xu et al. 2026). We compare four policy rows under identical data, steps, and cadence:

- **Frozen**: no trainable adapter;
- **Full-layer LoRA**: all language-model Q/K/V/O projections (Hu et al. 2022), always active;
- **Ungated KV**: the same last-eight layers, K/V targets, rank, and alpha as HGKV but active on all tokens;

B	Recent baseline	Replacement arms	CausalCache
1	Recent-1	prev. frame $\rightarrow$ archived	Selected-1
2	Recent-2	oldest slot $\rightarrow$ archived	Selected-$\leq$2
4	Recent-4	oldest slot $\rightarrow$ archived	Selected-$\leq$4
8	Recent-8	(built, not trained)	Selected-$\leq$8

Table 1: The fixed-budget design. Training arms replace the oldest recent slot with one archived frame (k=1); the deployed selector chooses at most B events from the full history, and the realized k is reported as a statistic. All contrasts at a given B are matched in image count; cross-B comparisons are analysis only.

- **HGKV (ours)**: token-gated, structurally bypassed at B=0.

Memory baselines are Recent-B, Random, OCR/RGB similarity, and the learned budget-aware selector (STOP allowed); an oracle over the candidate pool bounds headroom. Every method receives the same summaries and the same image budget.

Zero-Shot Benchmarks and Metrics

The policy interface and selector, trained on desktop data only, are evaluated (i) on held-out desktop decision groups (offline, per-B), (ii) zero-shot on an uncontaminated OSWorld task roster (Xie et al. 2024) with official evaluator scores (tasks whose trajectories seeded the witness corpus are excluded from zero-shot claims and reported separately as in-domain diagnostics), and (iii) zero-shot on a cross-app mobile memory benchmark of 117 runnable tasks (62 cross-app memory candidates, 55 single-app controls), which contributes no data to any training or selection decision. **[Pending: Fix the MobileWorld citation and the final roster hashes.]** Offline metrics: mean target log-likelihood margins, full-action equivalence (type, arguments, coordinates at τ=25 of a $[0, 999]$ grid; $\tau \in \{5, 10\}$ as sensitivity), repair and amplification rates, wrong-history harm, parser validity, and restored-image count. Closed-loop metrics: paired task success with cluster bootstrap, step counts, and wall time.

Results

Selectivity Is Not Inherited from the Frozen Policy

On the desktop development split at B=1 (94 groups, 77 trajectories), the choice of recent baseline matters. Against a *redundant* baseline that restores a byte-identical copy of the current screen (the deployment recent-selector's behavior), the frozen policy prefers the target-relevant archived frame by $+0.0333$ nats/token (95% CI $[+0.0118, +0.0567]$). Against the *informative* true previous frame, however, its preference vanishes: $+0.0013$ $[-0.0242, +0.0253]$, not significant (94/94 probes have a previous frame that differs from the current screen). To the frozen policy, a task-relevant archived frame and the recent frame it would replace are worth about the same; profitable reallocation must be learned. **[Pending: Insert the per-B frozen replacement effects and the $Q_B(k)$ curve with candidate-pool oracles, using the distinct-frame recent definition throughout.]**

Adapter (selected step)	DiD select	$\lvert A_r \rvert$	Wrong drift	Gates
HGKV (s150)	$\mathbf{+0.0111}$	0.0087	0.0085	pass
Ungated KV (s125)	$+0.0076$	0.0090	0.0063	pass
Full-layer LoRA (best)	$+0.0163$	0.0215	0.0116	fail

Table 2: Desktop DiD gate on 94 development groups (B=1; 2,000 episode-clustered bootstrap replicates). DiD select = $(A_s - A_r)$; caps require $|A_r|, |A_n| < 0.02$. HGKV's DiD CI is $[+0.0074, +0.0150]$; full-layer LoRA has no checkpoint satisfying the caps. HGKV B0 parity is bitwise exact.

Selective Use Within a Drift Envelope

Table 2 reports the pre-registered desktop DiD gate at B=1. All three adapters are trained for 150 matched steps; checkpoints are selected by the frozen rule (all gates passed, maximal DiD selection, earliest tie-break). HGKV passes every gate from step 50 onward: its selected checkpoint improves DiD selection to $+0.0111$ $[+0.0074, +0.0150]$ against the redundant-duplicate recent control, and a rescoring probe confirms $+0.0095$ $[+0.0056, +0.0138]$ (60/94 groups) against the informative true-previous-frame control, while holding recent-only and wrong-history drift well inside ϵ=0.02; its no-history path is bitwise identical to the frozen policy under randomized nonzero adapter weights. The layer- and rank-matched ungated control also learns selectivity ($+0.0076$), confirming that the last-eight K/V position carries signal, but token gating adds a paired per-group advantage of $+0.0035$ (95% CI $[+0.0007, +0.0062]$, 56/94 groups). Full-layer LoRA attains the largest raw selectivity yet fails the envelope at every checkpoint ($|A_r|$ between 0.021 and 0.027; wrong drift up to 0.029): it can select, but cannot stay anchored. Gating is what buys selectivity *inside* the envelope.

Reallocation Curves and Zero-Shot Transfer

[Pending: Report per-B replacement gates for $B \in \{1, 2, 4\}$ with the B=8 out-of-training extension, the $Q_B(k)$ reallocation curve and realized k per budget; then the zero-shot tables: on the uncontaminated OSWorld roster and the 117-task mobile memory benchmark, compare Frozen/HGKV $\times$ Recent-B/selector at matched budgets, with paired cluster bootstrap. The mobile benchmark contributes no training or selection signal.]

Ablations and Analyses

Planned ablations follow the fixed-budget factorization. Policy-side: per-B stratified DiD gates for HGKV, ungated KV, and full-layer LoRA (a pooled mean may hide help at B=4 and failure at B=1); the B=8 extrapolation; adapter depth and rank; removing the gain, content, or cap terms. Selector-side: recency-membership feature removal; candidate-pool size; STOP calibration; top-k versus bottom-k replacement; wrong-event and summary–image mismatch controls; a recent-memory dose analysis (varying window size without replacement) as an appendix. We also stratify by trajectory length, application, and positive-frame age.

Fixed-budget comparison (offline desktop heldout; primary setting $B = 4$)							
Policy	Allocation	B=1	B=2	B=4	B=8[†]	Realized k	Notes
Frozen	Recent-B	TBD	TBD	TBD	TBD	0	
Frozen	oldest slot $\rightarrow$ archived	TBD	TBD	TBD	TBD	1	B=1: $\approx$Recent
HGKV	Recent-B	TBD	TBD	TBD	TBD	0	
HGKV	oldest slot $\rightarrow$ archived	TBD	TBD	TBD	TBD	1	gate: Tab. 2
HGKV	learned selector (at-most-B, STOP)	TBD	TBD	TBD	TBD	TBD	k measured

Table 3: Skeleton of the fixed-budget main table. Each cell is a matched-count contrast on the desktop heldout split; the learned selector may choose STOP and is charged the images it actually uses; [†]B=8 is excluded from training and reports budget extrapolation. Cross-B comparisons are analysis only.

Limitations and Claim Boundary

Our frozen claim is about allocation, not modality:

> Under a fixed high-fidelity visual-memory budget B, Recent-B always allocates all slots to the most recent observations. CausalCache instead selects at most B events from the full interaction history. Its benefit therefore comes from reallocating a bounded memory budget, rather than adding extra visual context.

The claim is *not* that old frames are usually better than recent ones; it is that for some decisions, the conditional value of a few specific archived events exceeds that of the recent frames they replace, and CausalCache learns to recognize those states and events. The teacher-forced likelihood surrogate must ultimately be judged by closed-loop success. "Causal" refers only to controlled restoration interventions holding the remaining prompt fixed, not environment-level structural causality. The desktop DiD evidence reported here instantiates B=1; its training-time recent arm followed the deployment selector, whose top slot duplicates the current screen, so we separately re-scored the selected checkpoint against the true previous frame and report both controls (the v3 corpus freezes recent slots as distinct frames and discloses the deployment duplication separately). The per-B gates, $Q_B(k)$ curves, and the mobile zero-shot evaluation are pending, and we cannot guarantee the pretrained base policy never saw related data.

Conclusion

CausalCache reframes visual memory for GUI agents as budget reallocation: under a fixed number of high-fidelity slots, decide which events from the full interaction history deserve them, with recency as one candidate allocation rather than a privileged default. A token-gated KV interface learns to consume reallocated evidence inside a pre-registered drift envelope—where full-layer adaptation cannot stay—and a budget-aware selector learns when replacing recent frames with archived events pays. All learning happens on desktop data; mobile benchmarks remain untouched for zero-shot evaluation. **[Pending: Update with final per-B gates, $Q_B(k)$, and zero-shot findings.]**

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